Chap #10 (DC-Circuits)

*) Numericals (Pg #222):-

Question #01:-

Data:

$E_{mf} = E = 12V$

$r = 0.5\,\Omega$

$I = ?$

Solution:

For Maximum Current
load resistance $= R = 0$

$$E = I(r+R)$$

$$E = I(r+0) \quad \therefore R = 0$$

$$E = Ir$$

$$I = \frac{E}{r}$$

$$= \frac{12}{0.5}$$

$$[I = 24\,Amp] \text{ Ans}$$

Question # 02:-

Data:

$l = 12m$
$A = 4 \times 10^{-7} m^2$
$R = 6\Omega$
$\rho = ?$

Solution:

$$R = \frac{\rho L}{A}$$

$$\rho = \frac{AR}{L}$$

$$\rho = \frac{4 \times 10^{-7} \times 6}{12}$$

$[\rho = 2 \times 10^{-7} \Omega m]$ Ans

Question # 03:-

Data:

$E_1 = 1.2V$

$l_1 = 40\ cm = 0.4m$

$l_2 = 74\ cm = 0.74m$

$E_2 = ?$

Solution:

For balanced potentiometer:

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$E_2 = \frac{E_1 \times l_2}{l_1}$$

$$E_2 = \frac{1.2 \times 0.74}{0.4}$$

$$[\ E_2 = 2.22V\ ] \text{ Ans}$$

Question # 04:-

Data: (Series)

$R_1 = 1\Omega$
$R_2 = 2\Omega$
$R_3 = 3\Omega$
i) $R_{Eq} = ?$

ii) If $V = 24V$
then $V_1 = ?$, $V_2 = ?$, $V_3 = ?$

Solution:

$$R_{Eq} = R_1 + R_2 + R_3$$
$$= 1 + 2 + 3$$
$$[\ R_{Eq} = 6\Omega\ ]$$

$\therefore$ First find I

$$I = \frac{V}{R}$$
$$= \frac{24}{6}$$
$$[\ I = 4A\ ]$$

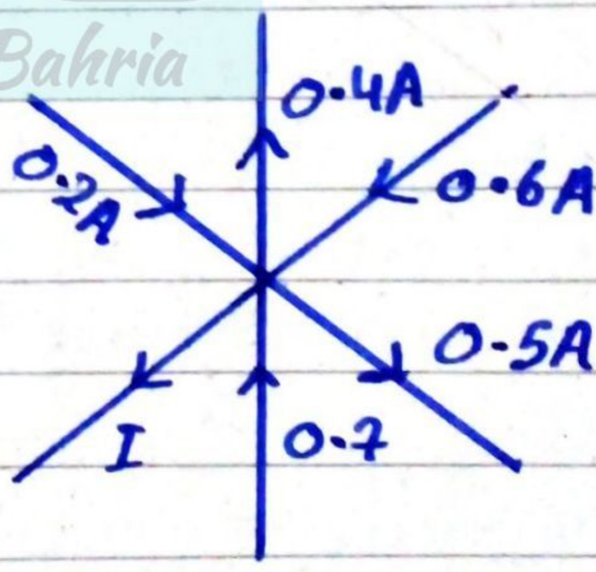
$\therefore I_1 = I_2 = I_3 = I = 4A$ (Series)

$V_1 = I_1 R_1 = 4 \times 1 = 4V$
$V_2 = I_2 R_2 = 4 \times 2 = 8V$
$V_3 = I_3 R_3 = 4 \times 3 = 12V$

Answer:

i) $R_{Eq} = 6\Omega$
ii) $V_1 = 4V$
$V_2 = 8V$
$V_3 = 12V$

Question #05:-



Data:

$I = ?$

Solution:

According to Kirchoff's law:

$$\Sigma I_{in} = \Sigma I_{out}$$

$$0.2+0.7+0.6 = I + 0.5 + 0.4$$
$$1.5 = I + 0.9$$
$$I = 1.5 - 0.9$$
$$[\, I = 0.6A \,] \text{ Ans}$$

Question # 06:-

Data:

$R_1 = 15\,\Omega$

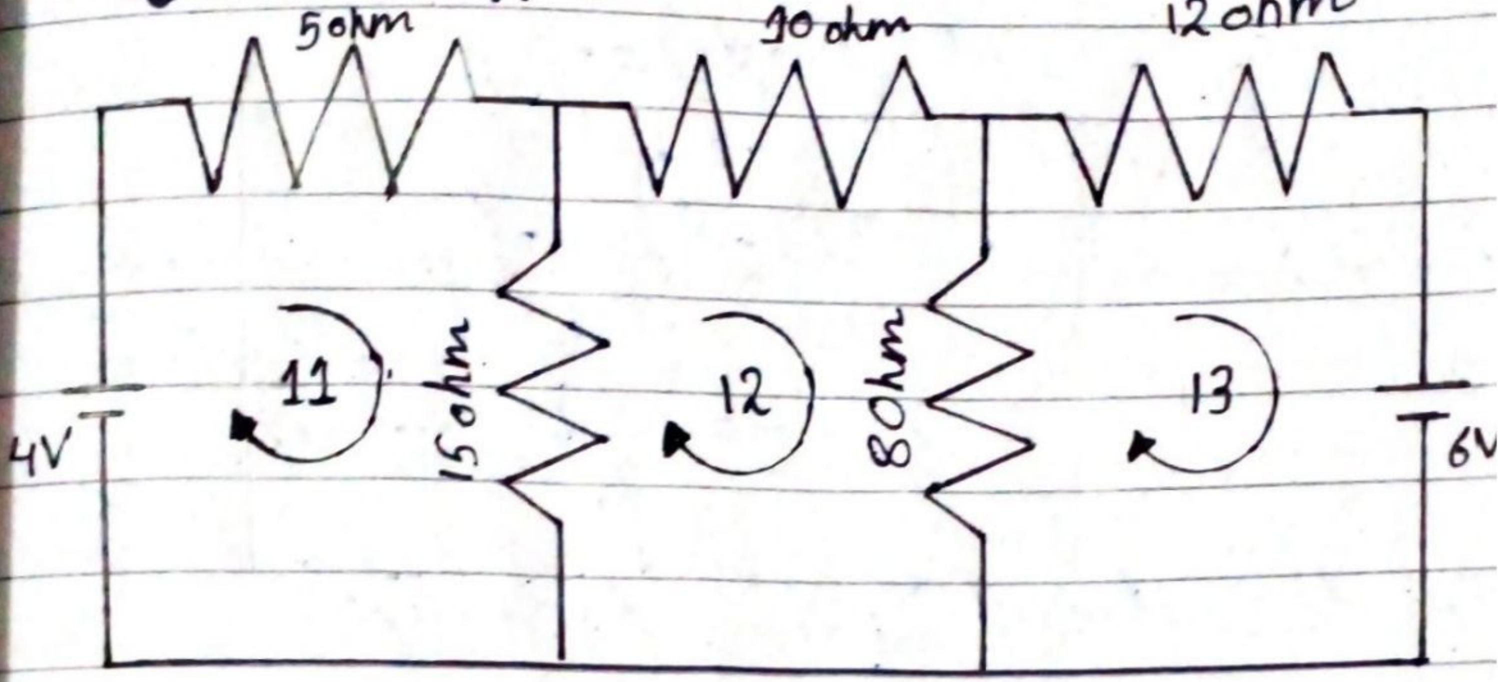
$\frac{r_2}{r_1} = \frac{5}{3}$

$R_2 = ?$

Solution:

$$\frac{R_2}{R_1} = \frac{l_2}{l_1}$$
$$\frac{R_2}{15} = \frac{5}{3}$$
$$R_2 = \frac{5 \times 15}{3}$$
$$[\, R_2 = 25\,\Omega \,] \text{ Ans}$$

Question # 07:-



Current through $10\,\Omega$ Resistance $(I_2) = ?$

According to Kirchoff's law on loop # 01:-

$$5I_1 + 15(I_1 - I_2) = 4$$

$$5I_1 + 15I_1 - 15I_2 = 4$$

$$20I_1 - 15I_2 = 4$$

$$20I_1 = 4 + 15I_2$$

$$I_1 = \frac{4 + 15I_2}{20} \quad \ldots (i)$$

For loop # 02:-

$$10I_2 + 8(I_2 - I_3) + 15(I_2 - I_1) = 0$$

$$10I_2 + 8I_2 - 8I_3 + 15I_2 - 15I_1 = 0$$

$$33I_2 - 15I_1 - 8I_3 = 0 \quad \ldots (ii)$$

For loop 3:-

$$12I_3+8(I_3-I_2)=-6$$

$$12I_3+8I_3-8I_2=-6$$

$$20I_3-8I_2=-6$$

$$20I_3=-6+8I_2$$

$$I_3=\frac{-6+8I_2}{20} \quad \ldots(iii)$$

Put the values of 'I_1' & 'I_3' in eq (ii):-

$$33I_2-15I_1-8I_3=0$$

$$33I_2-15\left(\frac{4+15I_2}{20}\right)-8\left(\frac{-6+8I_2}{20}\right)=0$$

$$\frac{660I_2-60-225I_2+48-64I_2}{20}=0$$

$$660I_2-225I_2-64I_2-60+48=0$$

$$371I_2-12=0$$

$$371I_2=12$$

$$I_2=\frac{12}{371}$$

$$[I_2=0.0323A(\text{clockwise})]$$